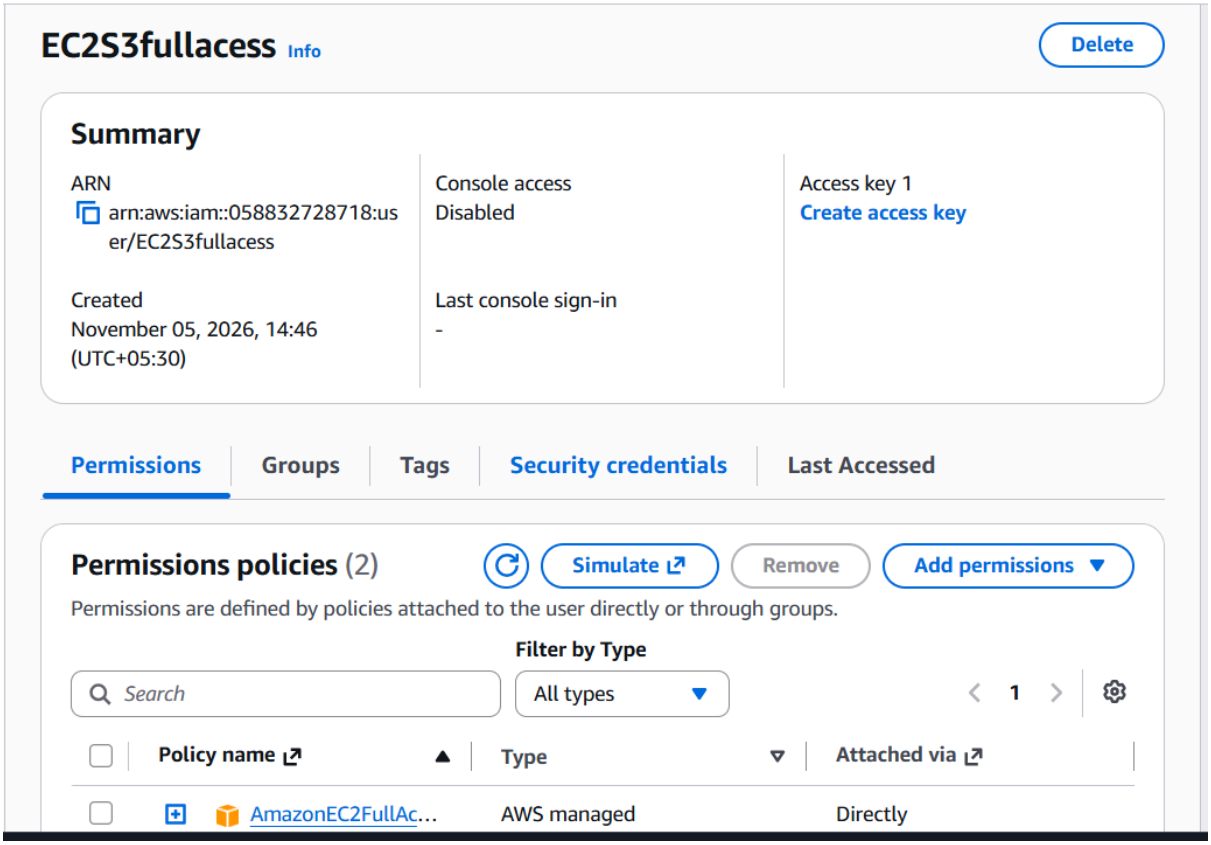
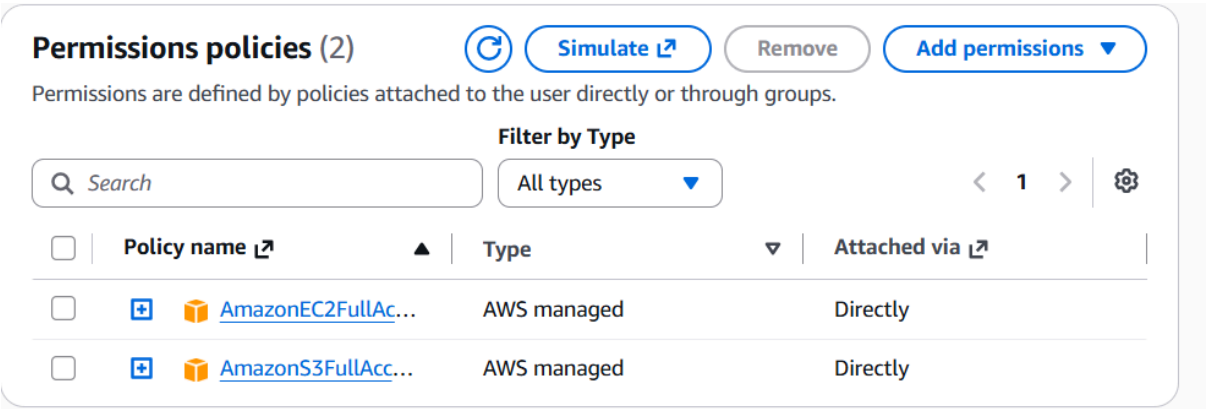
1. **Create one IAM user and assign EC2 and S3 full access roles.**

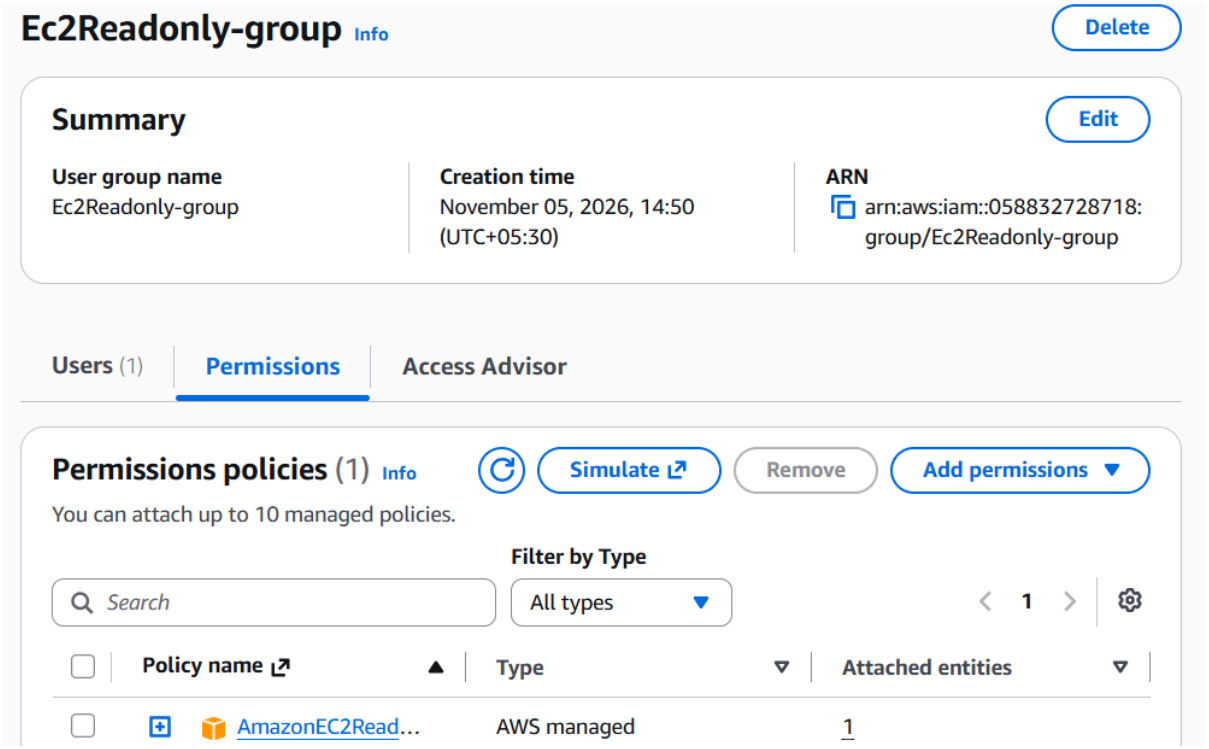




1. Open AWS Console → IAM.
2. Select Users → Create user.
3. Enter username: EC2S3fullaccess.
4. Click Next.
5. Select Attach policies directly.
6. Search and select AmazonEC2FullAccess.
7. Search and select AmazonS3FullAccess.
8. Click Next.

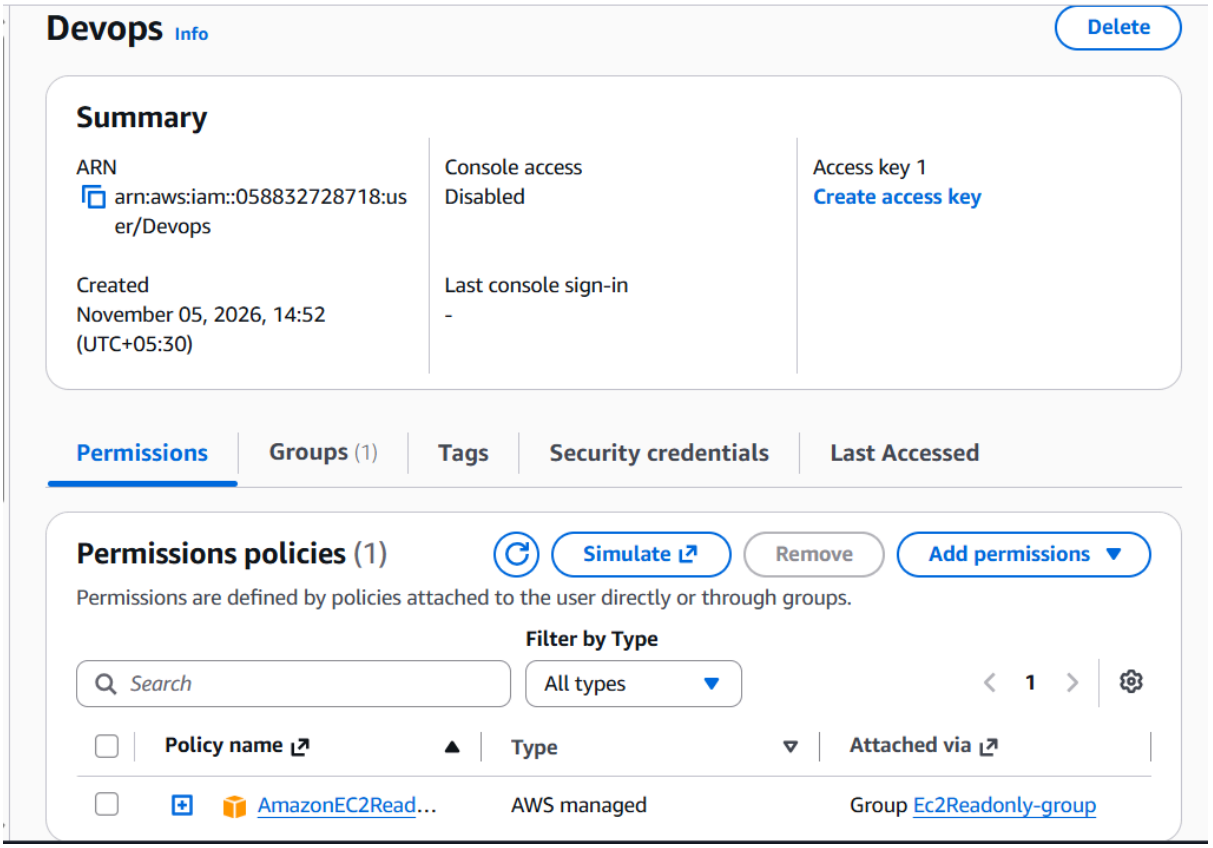
9. Review the permissions.
10. Click Create user.
11. Verify that both policies are attached under the Permissions tab.

2. Create one group in IAM and assign read access for EC2.



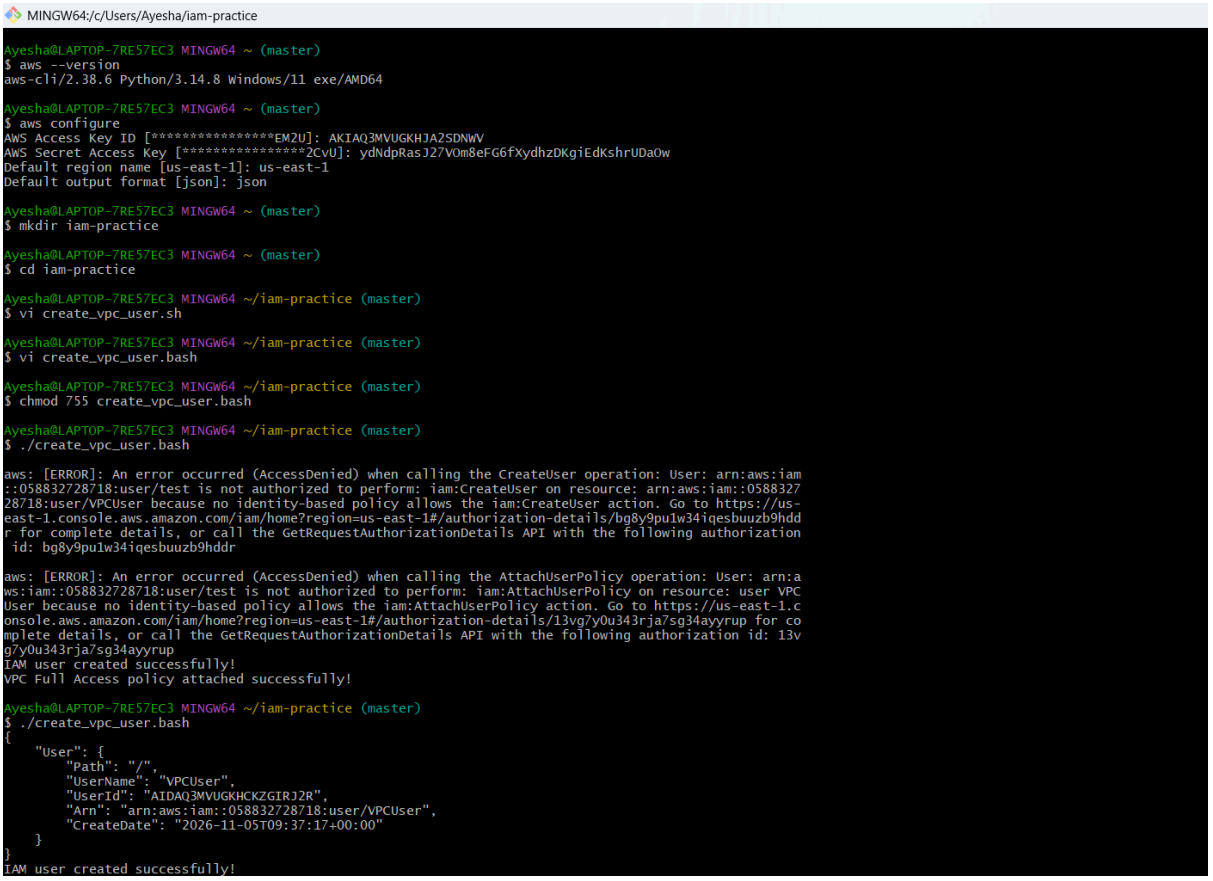
1. Open AWS Console → IAM.
2. Click User groups.
3. Click Create group.
4. Enter group name: Ec2Readonly-group.
5. Under Permissions policies, search for AmazonEC2ReadOnlyAccess.
6. Select the checkbox.
7. Click Create user group.
8. Open the newly created group.
9. Go to the Users tab.
10. Click Add users.
11. Select your existing IAM user.
12. Click Add users.
13. Verify the user and policy are listed in the group.

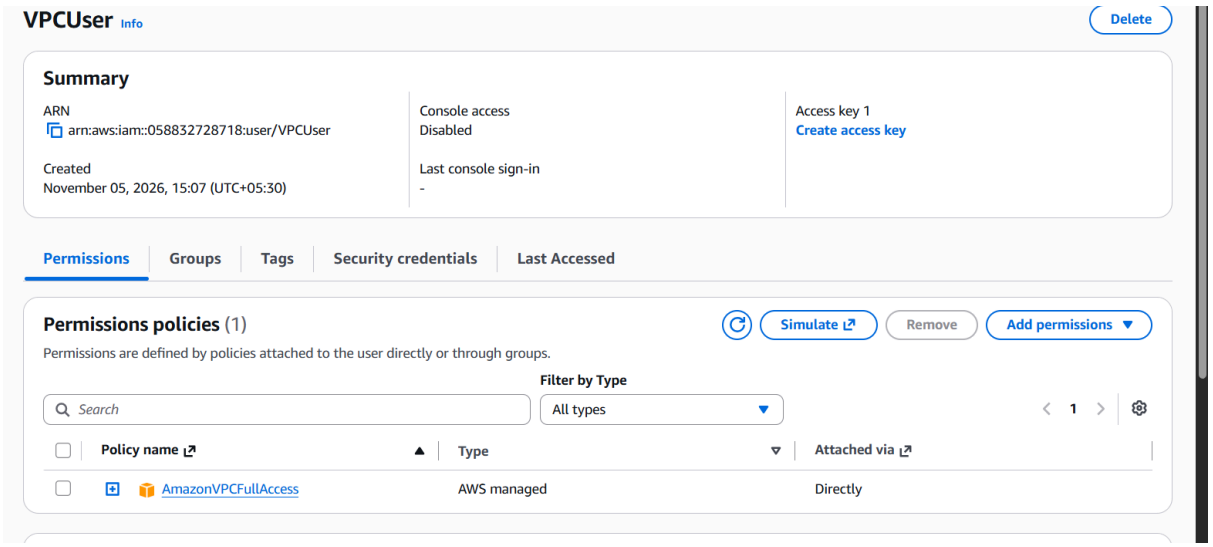
3.Create a new user named "Devops" and add to the group created in task 2.



1. Open AWS Console → IAM.
2. Click Users → Create user.
3. Enter username: Devops.
4. Click Next.
5. Under Permissions options, select Add user to group.
6. Select the existing group: Ec2Readonly-group.
7. Click Next.
8. Review the details.
9. Click Create user.
10. Open IAM → User groups → Ec2Readonly-group.
11. Click the Users tab.
12. Verify that Devops is listed in the group.

4. Write a bash script to create an IAM user with VPC full access.





Step 1: Open Git Bash

1. I opened Git Bash.
2. I checked AWS CLI using aws --version.
3. I configured AWS CLI using aws configure.
4. I verified AWS connectivity using aws sts get-caller-identity.

Step 2: Provide IAM permissions

1. I opened AWS Console → IAM → Users.
2. I selected the test user.
3. I opened Permissions → Add permissions.
4. I created an inline policy with iam:CreateUser and iam:AttachUserPolicy.
5. I saved the policy as IAMUserCreationPolicy.

Step 3: Create a Bash script

1. I created a folder using mkdir iam-practice.
2. I navigated into it using cd iam-practice.
3. I created a script using vi create_vpc_user.sh.

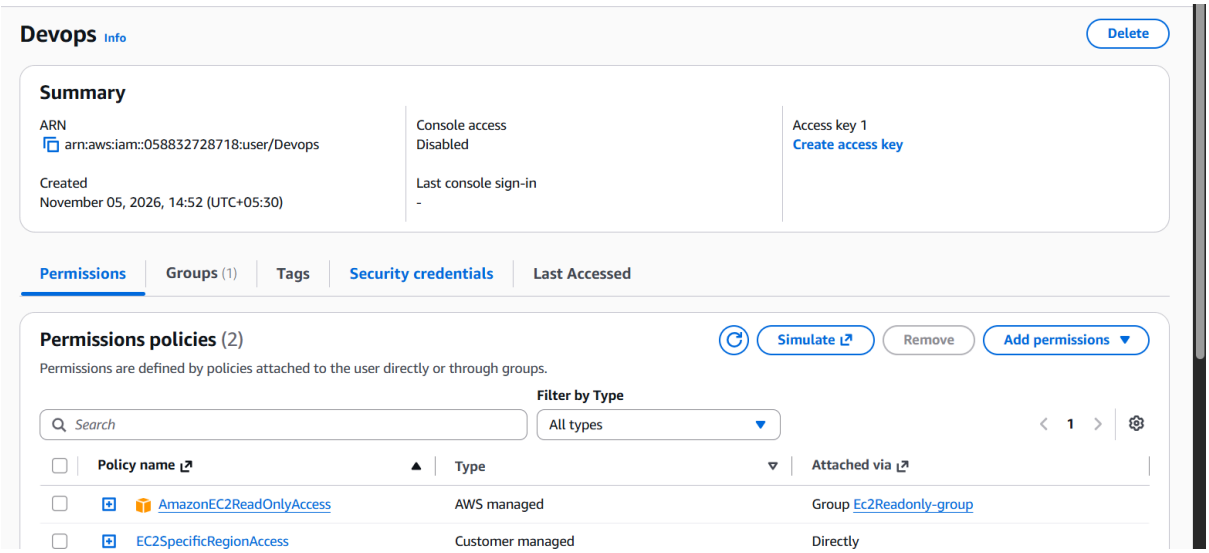
Step 4: Execute the script

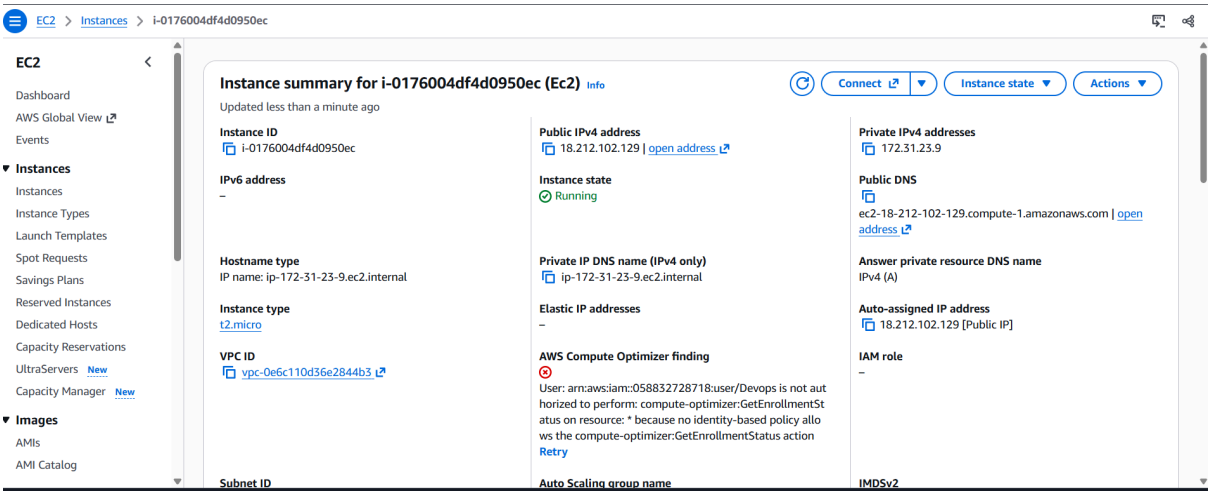
1. I saved the script using Esc → :wq.
2. I provided execution permission using chmod +x create_vpc_user.sh.
3. I executed the script using ./create_vpc_user.sh.

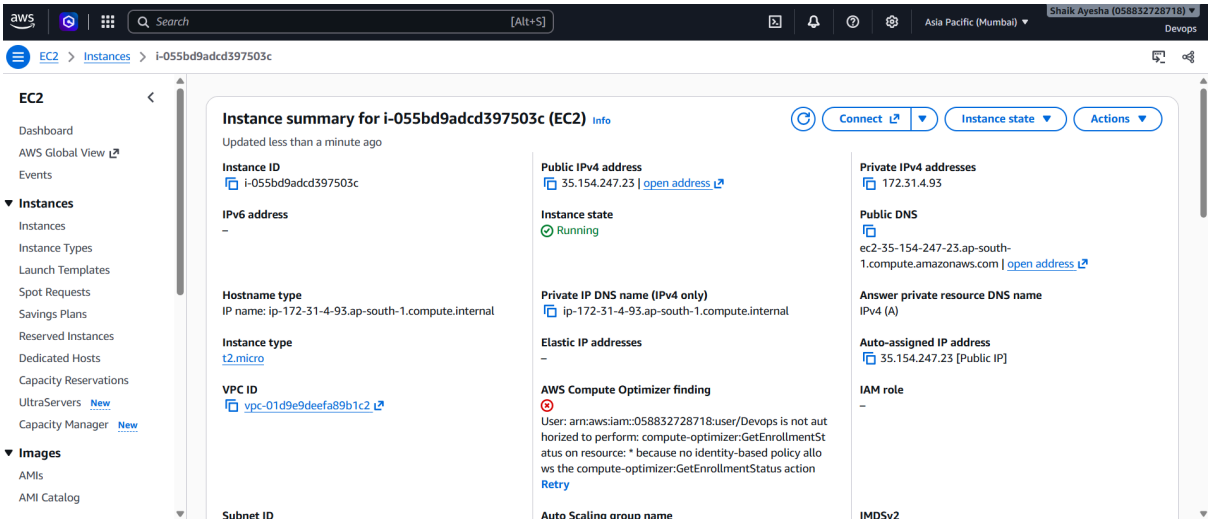
Step 5: Verify the result

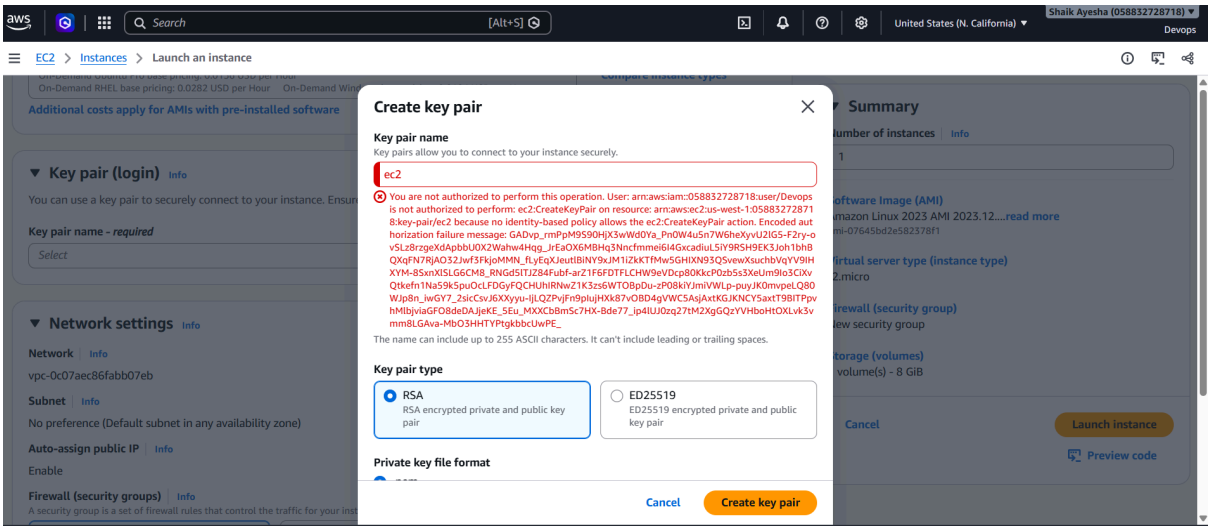
1. I opened AWS Console → IAM → Users.
2. I selected VPCUser.
3. I opened the Permissions tab.
4. I verified that AmazonVPCFullAccess was attached.

5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.




EC2 > Instances > i-0176004df4d0950ec
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instance summary for i-0176004df4d0950ec (Ec2) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0176004df4d0950ec
Public IPv4 address
18.212.102.129 | open address
Private IPv4 addresses
172.31.23.9
IPv6 address
–
Instance state
Running
Public DNS
ec2-18-212-102-129.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-23-9.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-23-9.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t2.micro
Elastic IP addresses
–
Auto-assigned IP address
18.212.102.129 [Public IP]
VPC ID
vpc-0e6c110d36e2844b3
AWS Compute Optimizer finding
User: arn:aws:iam::058832728718:user/Devops is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action
Retry
IAM role
–
Subnet ID
Auto Scaling group name
IMDSv2


aws
Search
[Alt+S]
Asia Pacific (Mumbai)
Shaik Ayesha (058832728718)
Devops
EC2 > Instances > i-055bd9adcd397503c
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instance summary for i-055bd9adcd397503c (EC2) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-055bd9adcd397503c
Public IPv4 address
35.154.247.23 | open address
Private IPv4 addresses
172.31.4.93
IPv6 address
–
Instance state
Running
Public DNS
ec2-35-154-247-23.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-4-93.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-4-93.ap-south-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t2.micro
Elastic IP addresses
–
Auto-assigned IP address
35.154.247.23 [Public IP]
VPC ID
vpc-01d9e9deefa89b1c2
AWS Compute Optimizer finding
User: arn:aws:iam::058832728718:user/Devops is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action
Retry
IAM role
–
Subnet ID
Auto Scaling group name
IMDSv2


aws
Search
[Alt+S]
United States (N. California)
Shaik Ayesha (058832728718)
Devops
EC2 > Instances > Launch an instance
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance.
Key pair name - required
Select
Network settings Info
Network Info
vpc-0c07aec86fabb07eb
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
ec2
You are not authorized to perform this operation. User: arn:aws:iam::058832728718:user/Devops is not authorized to perform: ec2:CreateKeyPair on resource: arn:aws:ec2:us-west-1:058832728718:key-pair/ec2 because no identity-based policy allows the ec2:CreateKeyPair action. Encoded authorization failure message: GADvp_rmPpM9S90HjX3wWd0Ya_Pn0W4u5n7W6heXyvU2IG5-F2ry-ovSLz8rzgeXdApbbU0X2Wahw4Hqg_JrEaOX6MBHq3Nncfmmei6l4GxcadiuL5iY9RSH9EK3Joh1bhBQXqFN7RjAO32Jwf3FkjoMMN_fLyEqXJeutlBiNY9xJM1iZkKTfMw5GHIXN93QSvewXsuchbVqYV9IHXYM-8SxnXlSLG6CM8_RNGd5ITJZ84Fubf-arZ1F6FDTFLCHW9eVDcp80KkcP0zb5s3XeUm9lo3CiXvQtkefn1Na59k5puOcLFDGyFQCHUhIRNwZ1K3zs6WTOBpDu-zP08kiYJmiVWLp-puyJK0mvpeLQ80WJp8n_iwGY7_2sicCsvJ6XXyyu-IjLQZPvjFn9plujHXk87vOBD4gVWC5AsjAxtKGJKNCYSaxtT9BITPpvhMIbjviaGFO8deDAJjeKE_5Eu_MXXCbBmSc7HX-Bde77_ip4IUJ0zq27tM2XgGQzYVHboHtOXLvk3vmm8LGAva-MbO3HHTYPtgkbbcUwPE_
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
Cancel
Create key pair
Summary
Number of instances Info
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

Task: Create an IAM Policy to Allow EC2 Access for a Specific User in Specific Regions Only

Step 1: Create IAM Policy

1. Log in to AWS Console using your administrator account.
2. Search for IAM.
3. Click Policies.
4. Click Create policy.
5. Select the JSON tab.
6. Paste the following policy:

```
None
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:RequestedRegion": [
            "us-east-1",
            "ap-south-1"
          ]
        }
      }
    }
```

```
  ]
}
```

Step 2: Save Policy

1. Click Next.
2. Enter policy name: EC2SpecificRegionAccess.
3. Click Create policy.

Step 3: Attach Policy to IAM User

1. Go to IAM → Users.
2. Select the existing user Devops.
3. Open Permissions.
4. Click Add permissions.
5. Select Attach policies directly.
6. Search for EC2SpecificRegionAccess.
7. Select the policy.
8. Click Next → Add permissions.

Step 4: Test Permissions

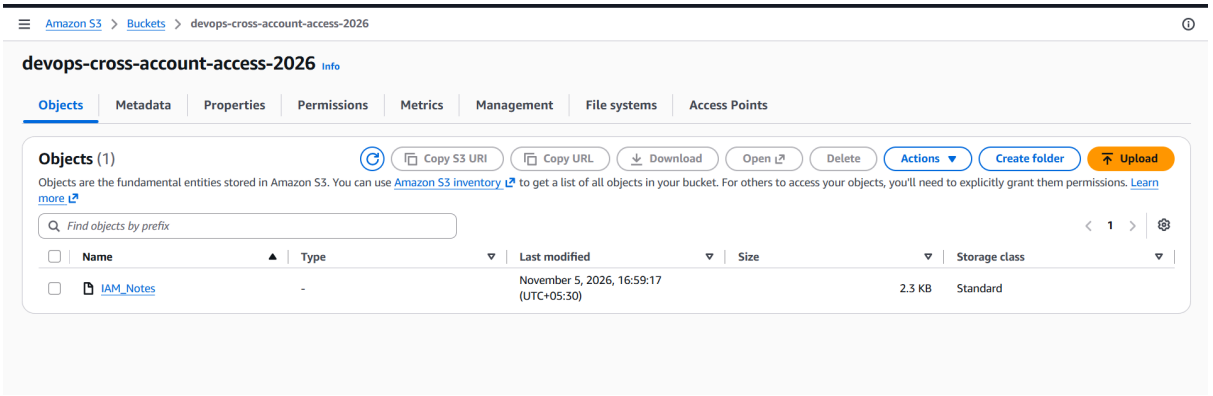
1. Log out of the administrator account.
2. Log in using the Devops IAM user.
3. Select us-east-1.
4. Open EC2 and test access.
5. Select ap-south-1.
6. Open EC2 and test access.
7. Select eu-west-1.
8. Test EC2 access.

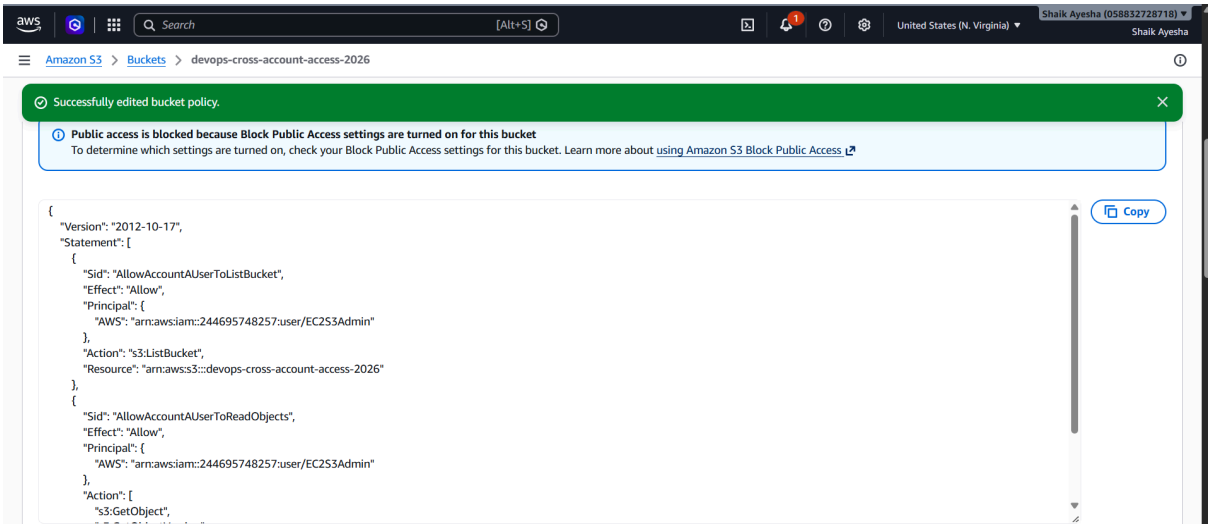
Expected result:

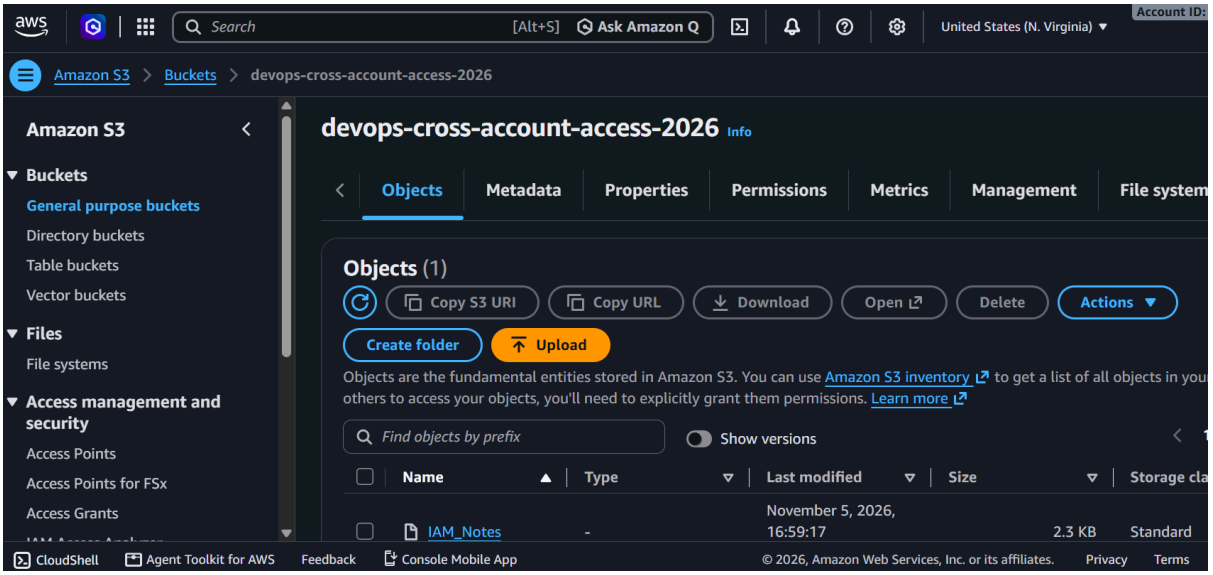
- us-east-1 – Allowed.
- ap-south-1 – Allowed.
- eu-west-1 – Denied, provided no other policy grants access.

6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

(Collaborate with a team member and execute this. This is mostly asked in every interview.)







1. My AWS Account

1. I logged in to AWS using my root account.
2. I opened the Amazon S3 service.
3. I created an S3 bucket named `devops-cross-account-access-2026`.
4. I opened the bucket and navigated to Permissions → Bucket policy.
5. I added a bucket policy allowing my friend's IAM user to list the bucket and read objects.
6. I specified my friend's AWS account ID and IAM username in the Principal ARN.
7. I saved the bucket policy.
8. I uploaded a sample file to the bucket.

2. My Friend's AWS Account

1. My friend logged in to AWS using their root account.
2. My friend opened the IAM service.
3. My friend created an IAM policy named `CrossAccountS3Access`.
4. My friend added permissions for `s3:ListBucket`, `s3:GetObject`, and `s3:GetObjectVersion`.
5. My friend specified my S3 bucket ARN and object ARN as resources.
6. My friend saved the IAM policy.
7. My friend created an IAM user named `EC2S3Admin`.
8. My friend attached the `CrossAccountS3Access` policy to the IAM user.

3. Testing Cross-Account Access

1. My friend logged out of the root account.
2. My friend logged in using the IAM user.
3. My friend opened the Amazon S3 service.
4. My friend accessed my bucket using its direct S3 bucket URL.
5. My friend verified that the bucket objects were visible.
6. My friend opened and downloaded the sample file to verify cross-account access.

4. Expected Result

- My friend can list my S3 bucket.
- My friend can view the objects inside the bucket.
- My friend can download objects.
- My friend cannot upload or delete objects because those permissions were not granted.